

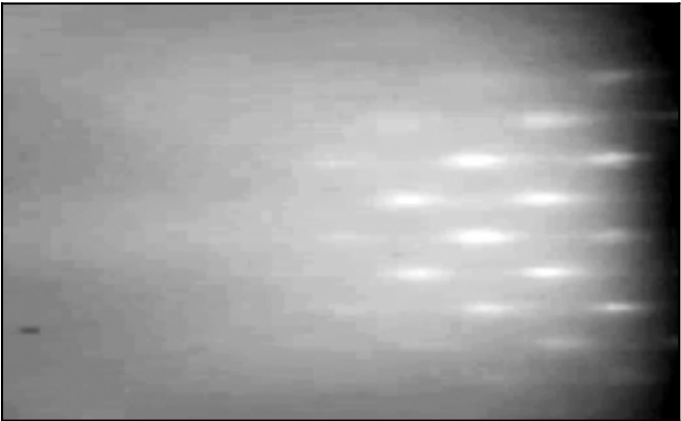
RHEED-conditioned AFM generation across measured roughness
fixed ordering, shared ± 8.3 nm color scale

RHEED key-frame ROI

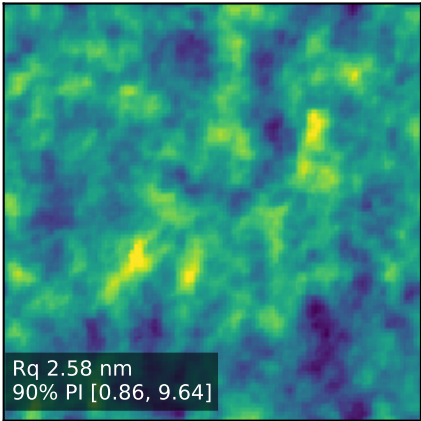
Generated AFM (nm)

Measured AFM (nm)

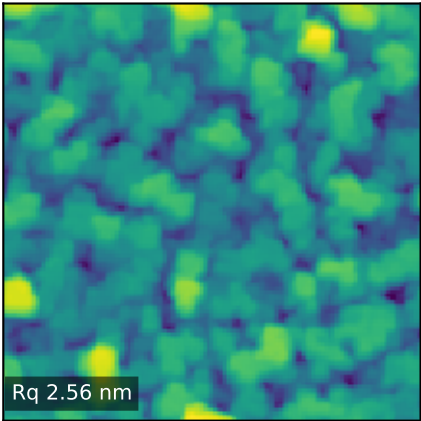
6029 [strict LOO]
confidence 20/100



Rq 2.58 nm
90% PI [0.86, 9.64]



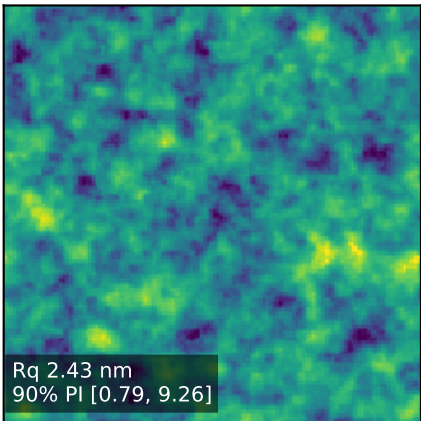
Rq 2.56 nm



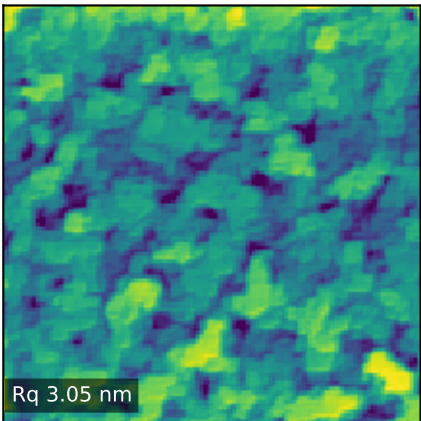
6070 [strict LOO]
confidence 11/100



Rq 2.43 nm
90% PI [0.79, 9.26]



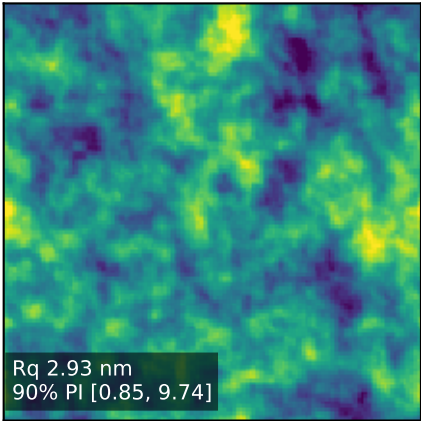
Rq 3.05 nm



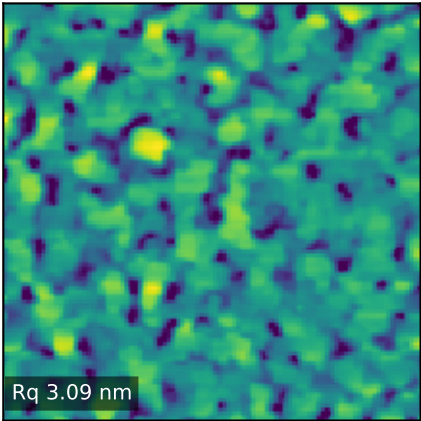
6062 [strict LOO]
confidence 17/100



Rq 2.93 nm
90% PI [0.85, 9.74]



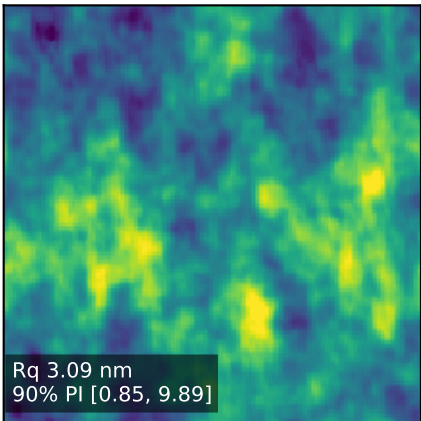
Rq 3.09 nm



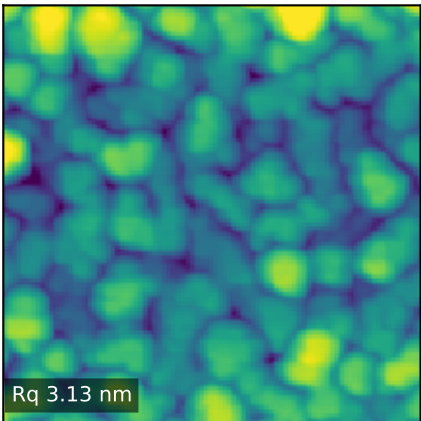
6085 [strict LOO]
confidence 20/100



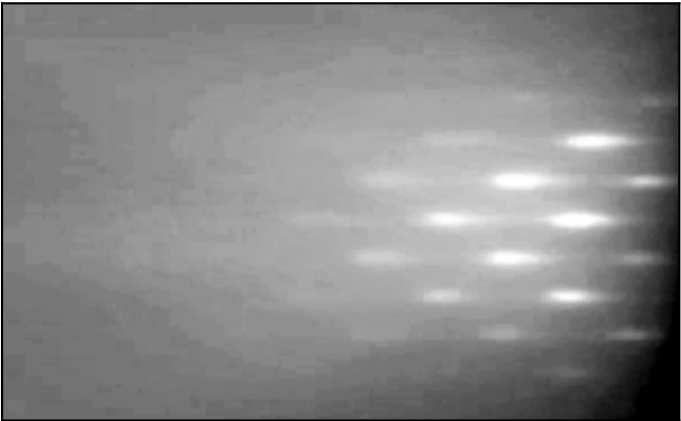
Rq 3.09 nm
90% PI [0.85, 9.89]



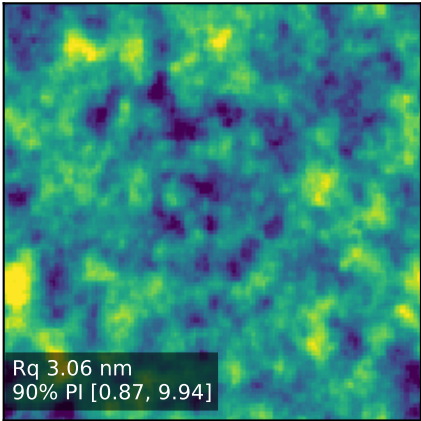
Rq 3.13 nm



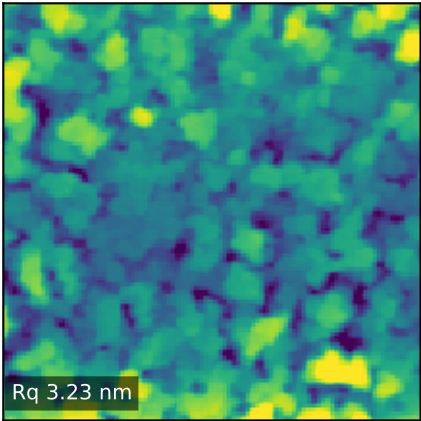
6056 [validation]
confidence 17/100



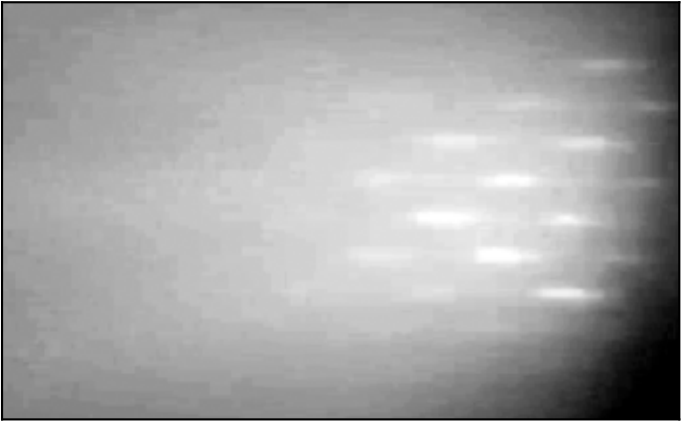
Rq 3.06 nm
90% PI [0.87, 9.94]



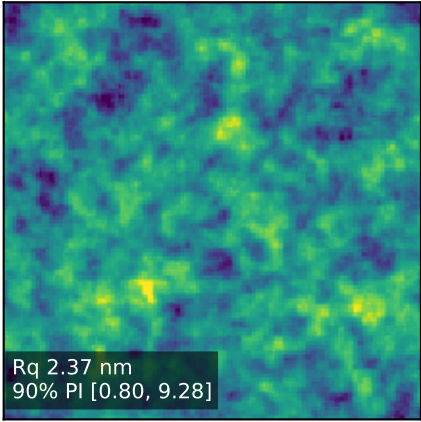
Rq 3.23 nm



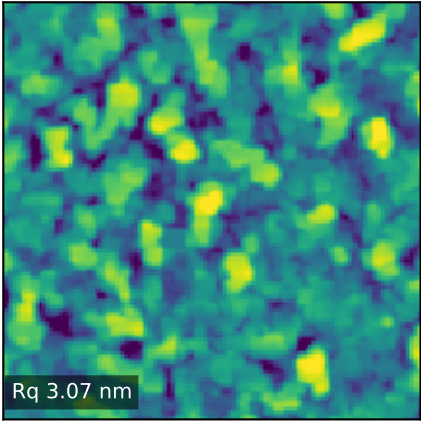
6047 [strict LOO]
confidence 19/100



Rq 2.37 nm
90% PI [0.80, 9.28]



Rq 3.07 nm



Training rows are strict leave-one-growth-group-out predictions; validation rows use the frozen training fit. Confidence is a conservative interval-width index, not a probability.